

LOKARAJ R

Artificial Intelligence and Machine Learning

Machine Learning Engineering Undergraduate Student at SRM Institute of Science and Technology. Actively trying to collaborate with teams to solve real-world problems to create meaningful impact. Managing end-to-end project workflows, including data analysis, model development, and deployment. Strong interest in innovative computing technologies and have a keen interest in collaborative learning and cross-cultural academic exchange, and ready to contribute to research and development, gaining experience in an international research environment.

WORK EXPERIENCE

Artificial Intelligence Intern

Sept 2024 - Dec 2024

Srishti Innovative Educational Services, Trivandrum

- Explored foundational AI concepts, analyzing data structures like tensors and how they are implemented with multi-dimensional data in neural networks with Tensorflow and Pytorch.
- Evaluated 5+ real-world case studies to find how AI is applied to solve specific problems in industries.
- **Recruiter notes:** Gained experience in ML workflow using Python, Jupyter, pandas, numpy, and scikit-learn for data analysis and preparation, and for building models.

PROJECTS

E-Learning Emotion Detection System

- Designed a system to detect and predict student emotions from live data streams for better engagement and built four unique datasets and appropriate models for real-time facial emotion recognition for performance and efficiency.

Real-Time Traffic Accident Prediction

- Developed a full project system architecture for real-time accident prediction and emergency response system, from data (CCTV) to model prediction and alert generation.

Video Surveillance in Google Cloud

- Created Python scripts using OpenCV for real-time motion detection in CCTVs and made an event ingestion pipeline to capture and log significant motions for immediate alerts in the cloud by GCP.

RESEARCH

A Comparative Analysis of YOLO Models in Vehicle Detection

First-Author, Paper ID:273 ICDMIS 2025

- YOLO (v8, v11,v12) models for real-time, license plate independent vehicle model identification, achieved over 97% accuracy on a custom Roboflow dataset for surveillance.

EDUCATION

Bachelor of Technology

June 2023 - May 2027

SRM University, Tiruchirappalli

Computer Science Engineering
specialization in Artificial Intelligence and
Machine Learning.

CONTACT

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SKILLS

Programming & Frameworks:

- Python
- Tensorflow
- Pytorch
- Scikit-learn
- OpenCV
- Pandas & Numpy
- Jupyter Notebooks
- SQL

Techniques:

- Computer Vision
- Object Detection (YOLO)
- Facial Emotion Recognition
- Predictive Modeling

OTHER

- Volunteered and developed a full-stack application with an ML churn Prediction model for Lions Club Ngo.
- Event Organizer – Neuro Tech Titans Club, Tech Shield Technical Quiz.

CERTIFICATES

- IBM Deep Learning with PyTorch, Keras, and Tensorflow
- AWS Data Engineering Virtual Internship
- Java Full Stack Developer Virtual Internship
- AWS AI-ML Virtual Internship
- AWS Cloud Virtual Internship